Multimodal Video Incident Identification User Guide

Goal:

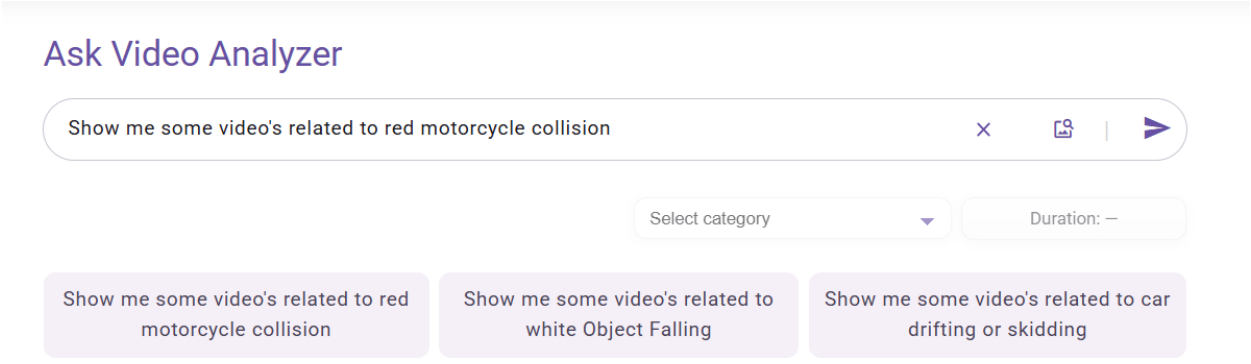
Enable **semantic, image-driven video retrieval** across a curated video dataset by converting both **query images** and **representative video keyframes** into a **shared multimodal embedding space (1408-dim)** using Vertex AI Multimodal Embeddings. The system allows analysts to upload an image such as an incident snapshot, traffic frame, product defect photo, or CCTV still and instantly retrieve the **most semantically similar video segments** from cloud storage. The solution integrates **Vertex AI**, **AlloyDB/pgvector**, and **GCS**, providing high-speed vector search.

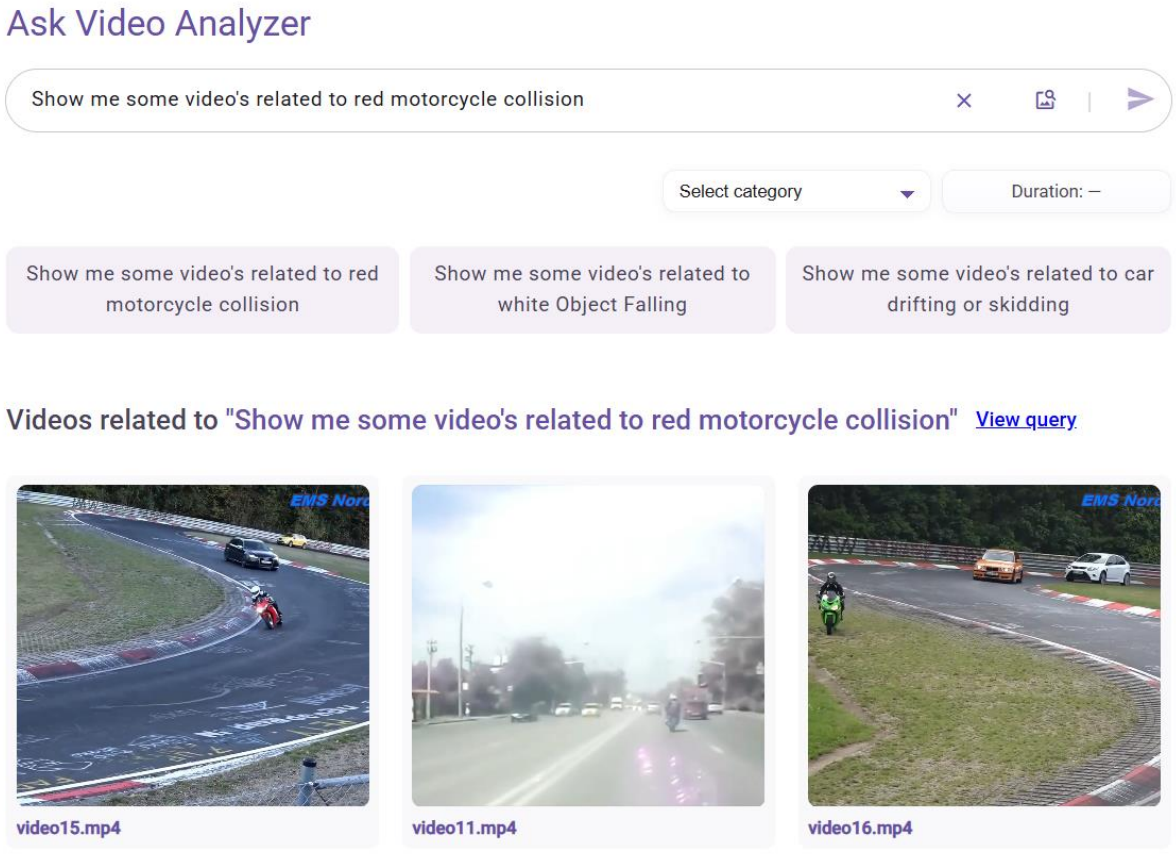
Workflow (Unchanged Semantics):

1. Ingestion trigger occurs when a new clip arrives in **GCS**.
2. A **Coordinator** orchestrates the sequence: Embed → Store → Index.
3. The **Vector Store** in **AlloyDB pgvector** saves the **VECTOR(1408),** video embedding plus metadata.
4. The **Retriever** converts a text query to an embedding and performs **ANN search**.
5. **MMR reranking** is applied to minimize near-duplicate results.
6. Results are returned with similarity scores, labels, and **GCS URIs**.

Example Query/Filters:

- Show me some video’s related to red motorcycle collision





Ask Video Analyzer

Show me some video's related to red motorcycle collision

Rear collision

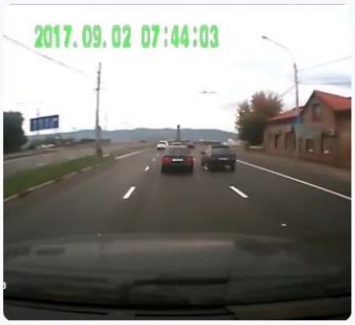
Duration: 4s

Show me some video's related to red motorcycle collision

Show me some video's related to white Object Falling

Show me some video's related to car drifting or skidding

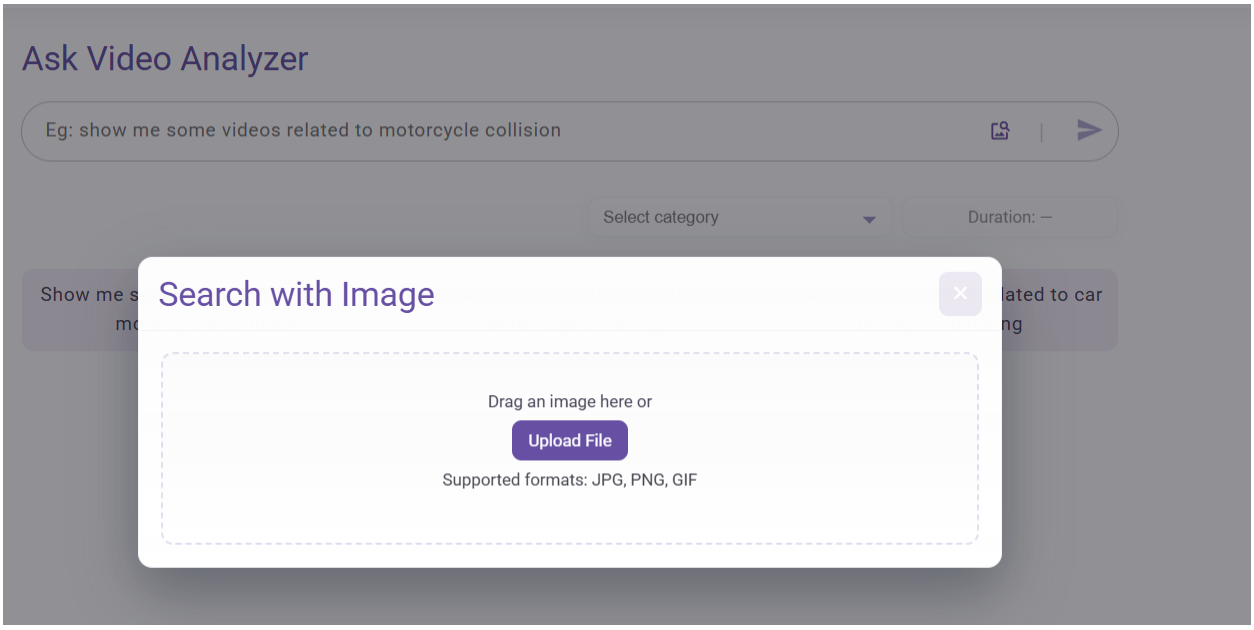
Videos related to "Show me some video's related to red motorcycle collision" View query



video14.mp4

video16.mp4

- Query using Image Search



Ask Video Analyzer

Image selected — search will use the image

Uploaded image: image (5).png

Remove

Select category

Duration: —

Show me some video's related to red motorcycle collision

Show me some video's related to white Object Falling

Show me some video's related to car drifting or skidding

Ask Video Analyzer

Image selected — search will use the image

Uploaded image: image (5).png

Remove

Select category

Duration: —

Show me some video's related to red motorcycle collision

Show me some video's related to white Object Falling

Show me some video's related to car drifting or skidding

Videos related to "Image search" View query

video18.mp4

video10.mp4

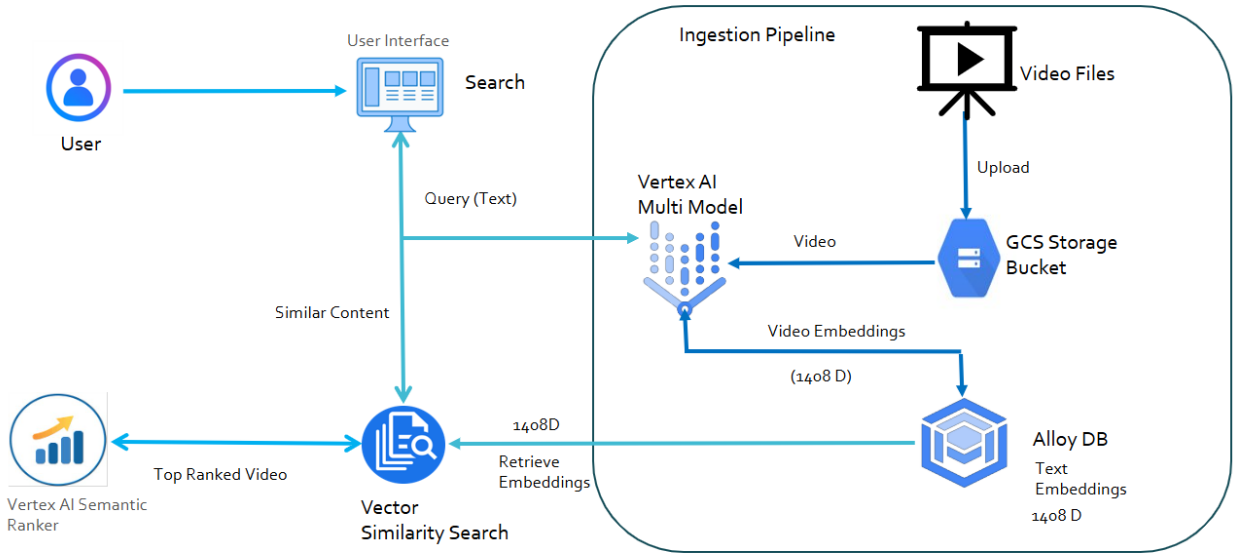
video8.mp4

Key Alloy DB Components:

- **AlloyDB for PostgreSQL** serves as the operational backbone, storing video metadata (id, label, duration_s, gcs_uri) along with embedding **VECTOR** fields.
- Maintain strict **embedding dimension** alignment by defining VECTOR sizes consistent with **Vertex AI Multimodal Embeddings** for video/text.
- **ANN Indexes (pgvector):**
    - Use **HNSW** or **IVFFlat** with the cosine operator class (e.g. **vector_cosine_ops**) and the <=> query operator.
    - Tune **IVFFlat** lists/probes or **HNSW** M / ef_search to balance latency and recall.
- **ScaNN** (optional, for large-scale deployments): Prefer **ScaNN indexes** with post-creation ANALYZE when working with very large corpora and if supported in your environment.

Technical Challenge & Business Impact:

Challenge:
Incident analysts must identify specific scenarios (e.g. **"night rollover"**) across numerous short clips. Manual review is slow and inconsistent, and simple **keyword tagging** fails to capture deeper **visual** and **temporal semantics**. At scale, basic search methods cannot provide **sub-second** retrieval or proper **auditability** (scores, parameters, reproducibility).

Impact:
A **multimodal**, **vector-search** workflow on **GCP** enables natural-language video search with rapid **ANN retrieval** and accurate **cross-modal matching** (text ↔ video).

Architecture Diagram